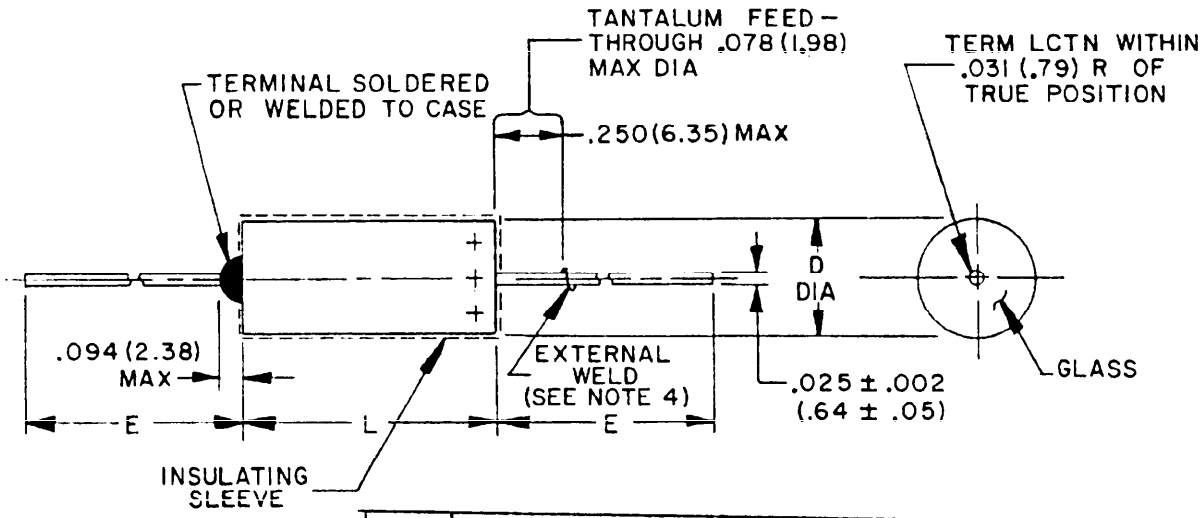
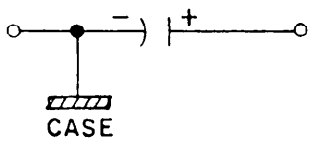
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR76

This specification is approved for use within the Department of the Air Force and is available for use by all Departments ments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-39006.





Case size	Dimensions			
	Basic case		Insulated case	
	L 1/	D	D max	E
	+.031 (.79) -.016 (.41)	±.016 (.41)		±.250 (6.35)
T1	.453 (11.51)	.188 (4.78)	.219 (5.56)	1.500 (38.10)
T2	.641 (16.28)	.281 (7.14)	.312 (7.92)	2.250 (57.15)
T3	.766 (19.46)	.375 (9.53)	.406 (10.31)	2.250 (57.15)
T4	1.062 (26.97)	.375 (9.53)	.406 (10.31)	2.250 (57.15)

1/ Length of basic case; sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest 0.01 mm) are given for general information only.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR76

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TABLE I. Capacitor characteristics.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			Case size	Part no. M39006/27-			
			+25°C	+85°C and +125°C					-55°C	+85°C	+125°C		Failure rate level (%1,000 hr)			
													M (1.0)	P (0.1)	R (0.01)	S (0.001)
volts, dc	µF	*percent	µA	µA	percent	volts,dc	volts,dc	ohms	percent	percent	percent					
6	220	20	2	9	50	4	6.9	36	-64	+13	+16	T1	0001	0089	0177	0265
6	220	10	2	9	50	4	6.9	36	-64	+13	+16	T1	0002	0090	0178	0266
6	820	20	3	14	155	4	6.9	18	-88	+16	+20	T2	0003	0091	0179	0267
6	820	10	3	14	155	4	6.9	18	-88	+16	+20	T2	0004	0092	0180	0268
6	1500	20	5	20	172	4	6.9	18	-90	+20	+25	T3	0005	0093	0181	0269
6	1500	10	5	20	172	4	6.9	18	-90	+20	+25	T3	0006	0094	0182	0270
6	2200	20	6	24	170	4	6.9	13	-90	+25	+30	T4	0007	0095	0183	0271
6	2200	10	6	24	170	4	6.9	13	-90	+25	+30	T4	0008	0096	0184	0272
8	180	20	2	9	41	5	6.9	45	-60	+13	+16	T1	0009	0097	0185	0273
8	180	10	2	9	41	5	9.2	45	-60	+13	+16	T1	0010	0098	0186	0274
8	680	20	3	14	130	5	9.2	22	-83	+16	+20	T2	0011	0099	0187	0275
8	680	10	3	14	130	5	9.2	22	-83	+16	+20	T2	0012	0100	0188	0276
8	1500	20	5	20	170	5	9.2	18	-90	+20	+25	T3	0013	0101	0189	0277
8	1500	10	5	20	170	5	9.2	18	-90	+20	+25	T3	0014	0102	0190	0278
8	1800	20	7	25	138	5	9.2	14	-90	+25	+30	T4	0015	0103	0191	0279
8	1800	10	7	25	138	5	9.2	14	-90	+25	+30	T4	0016	0104	0192	0280
10	150	20	2	9	34	7	11.5	54	-55	+13	+16	T1	0017	0105	0193	0281
10	150	10	2	9	34	7	11.5	54	-55	+13	+16	T1	0018	0106	0194	0282
10	560	20	3	16	106	7	11.5	27	-77	+16	+20	T2	0019	0107	0195	0283
10	560	10	3	16	106	7	11.5	27	-77	+16	+20	T2	0020	0108	0196	0284
10	1200	20	5	20	137	7	11.5	18	-88	+20	+25	T3	0021	0109	0197	0285
10	1200	10	5	20	137	7	11.5	18	-88	+20	+25	T3	0022	0110	0198	0286
10	1500	20	7	25	114	7	11.5	15	-88	+25	+30	T4	0023	0111	0199	0287
10	1500	10	7	25	114	7	11.5	15	-88	+25	+30	T4	0024	0112	0200	0288
15	100	20	2	9	30	10	17.2	72	-44	+13	+16	T1	0025	0113	0201	0289
15	100	10	2	9	30	10	17.2	72	-44	+13	+16	T1	0026	0114	0202	0290
15	390	20	3	16	74	10	17.2	31	-66	+16	+20	T2	0027	0115	0203	0291
15	390	10	3	16	74	10	17.2	31	-66	+16	+20	T2	0028	0116	0204	0292
15	820	20	6	24	111	10	17.2	22	-77	+20	+25	T3	0029	0117	0205	0293
15	820	10	6	24	111	10	17.2	22	-77	+20	+25	T3	0030	0118	0206	0294
15	1000	20	8	32	92	10	17.2	17	-77	+25	+30	T4	0031	0119	0207	0295
15	1000	10	8	32	92	10	17.2	17	-77	+25	+30	T4	0032	0120	0208	0296
25	68	20	2	9	22	15	28.8	90	-40	+12	+15	T1	0033	0121	0209	0297
25	68	10	2	9	22	15	28.8	90	-40	+12	+15	T1	0034	0122	0210	0298
25	270	20	3	16	55	15	28.8	33	-62	+13	+16	T2	0035	0123	0211	0299
25	270	10	3	16	55	15	28.8	33	-62	+13	+16	T2	0036	0124	0212	0300
25	560	20	7	28	76	15	28.8	24	-72	+20	+25	T3	0037	0125	0213	0301
25	560	10	7	28	76	15	28.8	24	-72	+20	+25	T3	0038	0126	0214	0302
25	680	20	8	32	63	15	28.8	19	-72	+25	+30	T4	0039	0127	0215	0303
25	680	10	8	32	63	15	28.8	19	-72	+25	+30	T4	0040	0128	0216	0304
30	56	20	2	9	22	20	34.5	100	-38	+12	+15	T1	0041	0129	0217	0305
30	56	10	2	9	22	20	34.5	100	-38	+12	+15	T1	0042	0130	0218	0306
30	220	20	3	16	42	20	34.5	36	-60	+13	+16	T2	0043	0131	0219	0307
30	220	10	3	16	42	20	34.5	36	-60	+13	+16	T2	0044	0132	0220	0308

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TABLE I. Capacitor characteristics - Continued.

Rated voltage (+85°C)	Cap.	Cap. tolerance	DC leakage (max)		Dissipation factor (max)	Derated voltage (+125°C)	Surge voltage (+85°C)	Impedance (max)	Capacitance change at			Case size	Part no. M39006/27-			
			+25°C	+85°C and +125°C					-55°C	+85°C	+125°C		Failure rate level (%1,000 hr)			
													M (1.0)	P (0.1)	R (0.01)	S (0.001)
volts, dc	µF	*percent	µA	µA	percent	volts,dc	volts,dc	ohms	percent	percent	percent					
30	470	20	8	32	64	20	34.5	25	-65	+20	+25	T3	0045	0133	0221	0309
30	470	10	8	32	64	20	34.5	25	-65	+20	+25	T3	0046	0134	0222	0310
30	560	20	9	36	55	20	34.5	20	-65	+25	+30	T4	0047	0135	0223	0311
30	560	10	9	36	55	20	34.5	20	-65	+25	+30	T4	0048	0136	0224	0312
50	33	20	2	9	12.3	30	57.5	135	-29	+10	+12	T1	0049	0137	0225	0313
50	33	10	2	9	12.3	30	57.5	135	-29	+10	+12	T1	0050	0138	0226	0314
50	120	20	4	24	22.5	30	57.5	49	-42	+12	+15	T2	0051	0139	0227	0315
50	120	10	4	24	22.5	30	57.5	49	-42	+12	+15	T2	0052	0140	0228	0316
50	270	20	8	32	37	30	57.5	29	-46	+20	+25	T3	0053	0141	0229	0317
50	270	10	8	32	37	30	57.5	29	-46	+20	+25	T3	0054	0142	0230	0318
50	330	20	9	36	38	30	57.5	22	-46	+25	+30	T4	0055	0143	0231	0319
50	330	10	9	36	38	30	57.5	22	-46	+25	+30	T4	0056	0144	0232	0320
60	27	20	3	12	10.2	40	69	144	-24	+10	+12	T1	0057	0145	0233	0321
60	27	10	3	12	10.2	40	69	144	-24	+10	+12	T1	0058	0146	0234	0322
60	100	20	4	20	19	40	69	54	-36	+12	+15	T2	0059	0147	0235	0323
60	100	10	4	20	19	40	69	54	-36	+12	+15	T2	0060	0148	0236	0324
60	220	20	8	32	30	40	69	29	-40	+16	+20	T3	0061	0149	0237	0325
60	220	10	8	32	30	40	69	29	-40	+16	+20	T3	0062	0150	0238	0326
60	270	20	9	36	27	40	69	23	-45	+20	+25	T4	0063	0151	0239	0327
60	270	10	9	36	27	40	69	23	-45	+20	+25	T4	0064	0152	0240	0328
75	22	20	3	12	8.5	50	86.2	157	-19	+10	+12	T1	0065	0153	0241	0329
75	22	10	3	12	8.5	50	86.2	157	-19	+10	+12	T1	0066	0154	0242	0330
75	82	20	4	24	15.2	50	86.2	63	-30	+12	+15	T2	0067	0155	0243	0331
75	82	10	4	24	15.2	50	86.2	63	-30	+12	+15	T2	0068	0156	0244	0332
75	180	20	9	36	24.4	50	86.2	30	-35	+16	+20	T3	0069	0157	0245	0333
75	180	10	9	36	24.4	50	86.2	30	-35	+16	+20	T3	0070	0158	0246	0334
75	220	20	10	40	37	50	86.2	24	-40	+20	+25	T4	0071	0159	0247	0335
75	220	10	10	40	37	50	86.2	24	-40	+20	+25	T4	0072	0160	0248	0336
100	10	20	3	12	4.5	65	115	200	-17	+10	+12	T1	0073	0161	0249	0376
100	10	10	3	12	4.5	65	115	200	-17	+10	+12	T1	0074	0162	0250	0338
100	39	20	5	24	10.4	65	115	80	-20	+12	+15	T2	0075	0163	0251	0339
100	39	10	5	24	10.4	65	115	80	-20	+12	+15	T2	0076	0164	0252	0340
100	68	20	10	40	11.3	65	115	40	-30	+14	+16	T3	0077	0165	0253	0341
100	68	10	10	40	11.3	65	115	40	-30	+14	+16	T3	0078	0166	0254	0342
100	120	20	12	48	25	65	115	30	-35	+15	+17	T4	0079	0167	0255	0343
100	120	10	12	48	25	65	115	30	-35	+15	+17	T4	0080	0168	0256	0344
125	6.8	20	3	12	6	85	144	300	-14	+10	+12	T1	0081	0169	0257	0345
125	6.8	10	3	12	6	85	144	300	-14	+10	+12	T1	0082	0170	0258	0346
125	27	20	5	24	7.2	85	144	90	-18	+12	+15	T2	0083	0171	0259	0347
125	27	10	5	24	7.2	85	144	90	-18	+12	+15	T2	0084	0172	0260	0348
125	47	20	10	40	7.9	85	144	50	-26	+14	+16	T3	0085	0173	0261	0349
125	47	10	10	40	7.9	85	144	50	-26	+14	+16	T3	0086	0174	0262	0350
125	82	20	12	80	17.4	85	144	32	-30	+15	+17	T4	0087	0175	0263	0351
125	82	10	12	80	17.4	85	144	32	-30	+15	+17	T4	0088	0176	0264	0352

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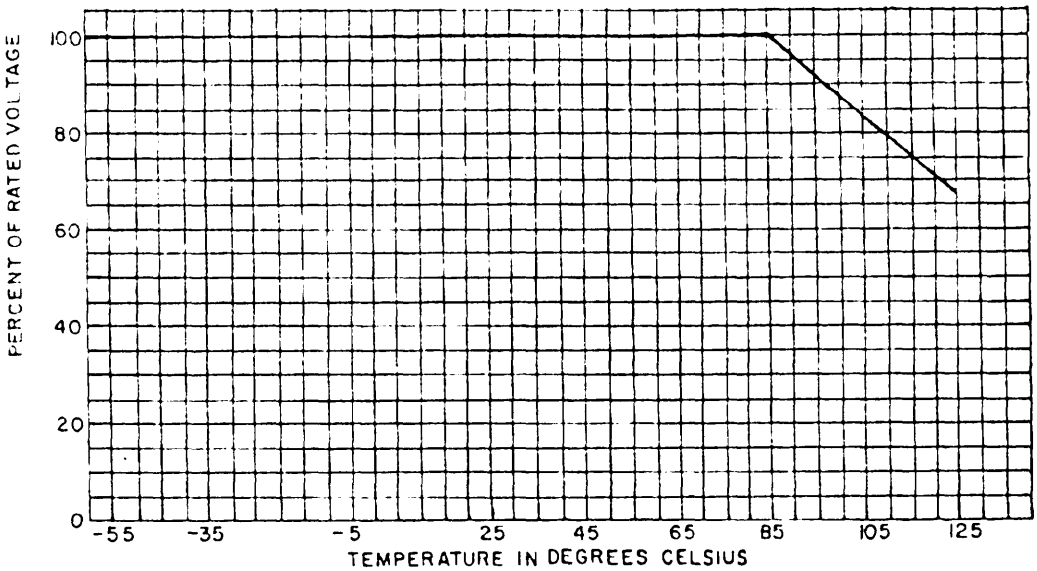


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular, insulated, bonded metal with the innermost layer being tantalum and comprising a minimum of 10 percent of the total wall.

Seal type: Hermetic (glass-to-tantalum anode).

Terminals:

Axial-wire lead: Anode (type N4 or N5 of MIL-STD-1276). Cathode (type C4 or C5 of MIL-STD-1276).

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (Cap.) (nominal): See table I.

Capacitance tolerance: See table I.

Failure rate level: See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage (DCL) (max) (at +25°C): See table I.

Capacitance (nominal): Method 305 of MIL-STD-202 (see table I).

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 g's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 g's).

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Thermal shock: Method 107 of MIL-STD-202, condition A.

DCL: 125 percent (max) of +25°C value of table I.

Δ Cap: Within ±5 percent of initial measurement.

DF: 115 percent of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hours).

Solderability: Method 208 of MIL-STD-202.

Terminal strength: Pull, method 211 of MIL-STD-202, condition A.

Wire-lead-bend: In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.

DCL: 125 percent (max) of +25°C value of table I.

ΔCap: Within ±8 percent of initial measurement.

DF: 115 percent (max) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202. 2,000 V dc minimum.

Insulation resistance (IR) method 302 of MIL-STD-202, condition B. 100 megohms minimum.

Low temperature (storage): Method 502 of MIL-STD-810.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Stability at low and high temperatures:

Step 1 (+25°C):

DCL: See table I.

Cap: Within tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

Imp.: See table I.

ΔCap: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

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Step 4 (+85°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 5 (+125°C):

DCL: See table I.

ΔCap: See table I.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap: Within ±5 percent of step 1 value.

DF: See table I.

Reverse voltage: Capacitors shall withstand a dc potential of 1 volt, applied in the reverse direction, for 125 ±10 hours. The ambient temperature during the test shall be +85°C. Capacitors shall be maintained at +85°C and dc rated voltage shall be applied in the forward direction for 125 ±10 hours. Capacitors shall then be returned to room ambient temperature and the dc leakage, capacitance, and dissipation factor shall be measured as specified in paragraphs 4.7.3, 4.7.4, and 4.7.5 respectively, of MIL-C-39006.

DCL: See table I.

ΔCap: Within ±10 percent of initial measurement.

DF: See table I.

Life at +85°C: Method 108 of MIL-STD-202. 2,000 hour (qualification):

DCL: See table I.

ΔCap: Within +10 to -25 percent of initial measurement for 6 volt units;
Within +10 to -20 percent of initial measurement for 8- and 10- volt units;
Within +10 to -15 percent of initial measurement for 15 volt units;
Within ±10 percent of initial measurement for all other units.

DF: See table I.

10,000 hour (group B) and extended life:

DCL: (+85°C), 125 percent (max) of table I value.

DCL: (+25°C), See table I.

ΔCap: Within +10 percent, -60 percent of initial measurement.

DF: 200 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

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Vacuum life:

DCL: See table I.

ΔCap: Within ±20 percent of initial measurement.

DF: See table I.

Life at +125°C: Method 108 of MIL-STD-202.

DCL: See table I.

ΔCap: Within +10 to -25 percent of initial measurement.
For 6 volt units; within +10 to -20 percent of initial measurement for 8- and 10- volt units; within +10 to -15 percent of initial measurement for 15-volt units; within ±10 percent of initial measurements for all other units.

DF: See table I.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

10,000 hours (group B):

DCL: (+125°C), 125 percent (max) of table I value.

DCL: (+25°C), See table I.

ΔCap: Within +10 percent, -60 percent of initial measurement.

DF: 400 percent (max) of table I value.

DWV: 2,000 V dc, minimum.

IR: 100 megohms, minimum.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 feet).

Test voltage: Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

DCL: See table I.

ΔCap: Within ±5 percent of initial measurement.

DF: See table I.

Marking: In accordance with MIL-C-39006.

Polarity: Series of plus (+) symbols encircling positive end (see figure 1).

Voltage groups (+85°C rated voltage):

I - - - - - 6 to 15 V dc inclusive.

II - - - - - 25 to 75 V dc inclusive.

III - - - - - 100 and 125 V dc.

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